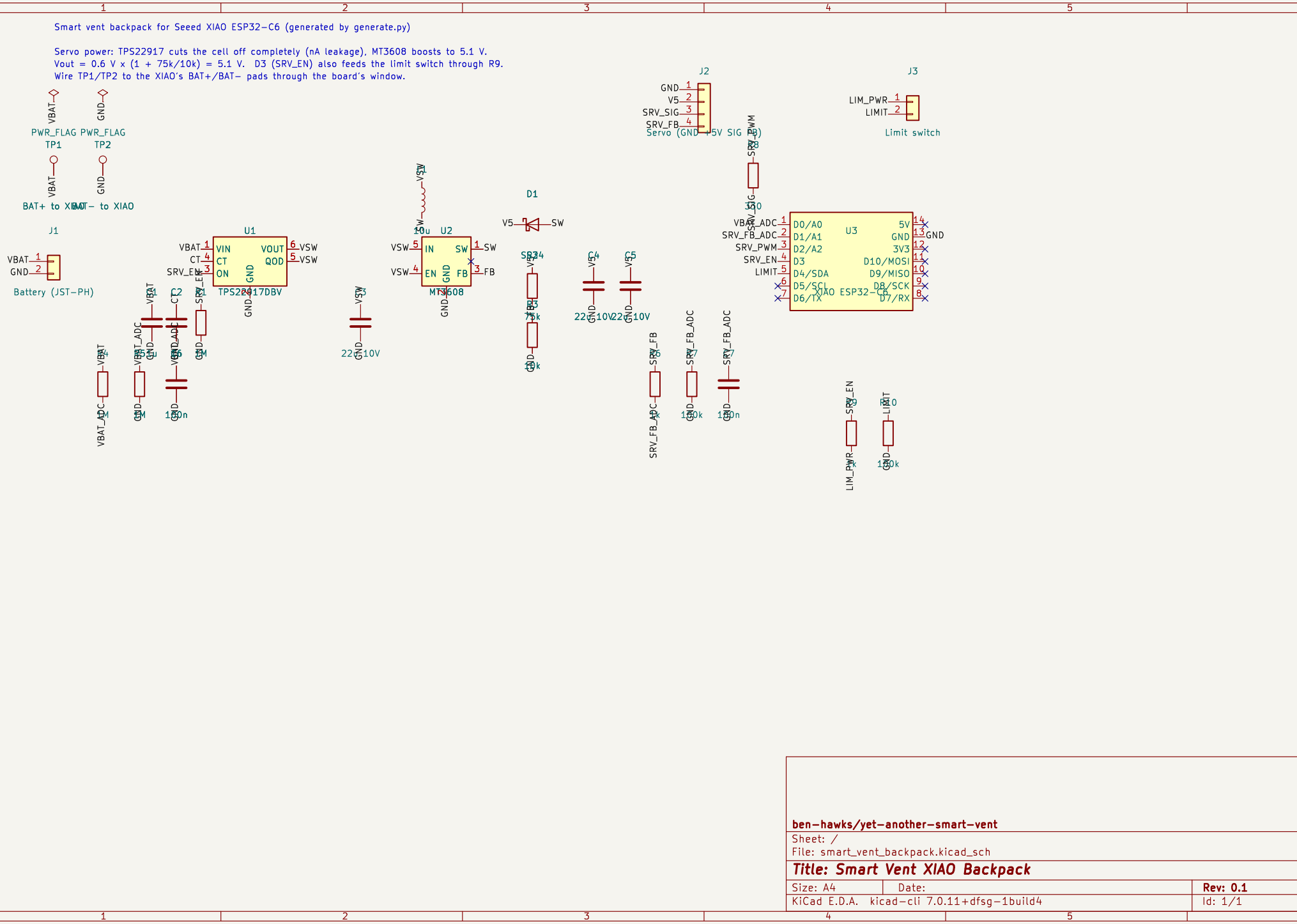

Smart vent backpack for Seeed XIAO ESP32-C6 (generated by generate.py)
Servo power: TPS22917 cuts the cell off completely (nA leakage), MT3608 boosts to 5.1 V.
Vout = 0.6 V x (1 + 75k/10k) = 5.1 V. D3 (SRV_EN) also feeds the limit switch through R9.
Wire TP1/TP2 to the XIAO's BAT+/BAT- pads through the board's window.
PWR_FLAG
TP1
TP2
BAT+ to XIAO
BAT- to XIAO
J1
Battery (JST-PH)
U1
TPS22917DBV
U2
MT3608
D1
SS34
J2
Servo (GND +5V SIG FB)
J3
Limit switch
U3
XIAO ESP32-C6
10u
22u 10V
100n
1M
100k
75k
10k
ben-hawks/yet-another-smart-vent
Sheet: /
File: smart_vent_backpack.kicad_sch
Title: Smart Vent XIAO Backpack
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1